

Q1. Define True/False. If false give the correct answer.

- All class requires a static main method.
- A constructor has the same name as the class in which it resides and syntactically similar to a method but does not have any return type.
- Private members of a class can be access from its subclass within same package.
- Abstract class cannot be inherited.
- All methods in an interface must be abstract.

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Q2: Check the following java program carefully. Do you think there is any compilation error? If there is find the errors, fix them and write the output. If there is no error then write only output.

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```
class A {
    int i, j;
    A(int a, int b) {
        System.out.println("A is 3rd.");
        i = a;
        j = b;
    }
    void show() {
        System.out.println("i and j: " + i + " " + j);
    }
}
class B extends A {
    int k;
    B(int a, int b, int c) {
        super(a, b);
        System.out.println("B is 2nd.");
        k = c;
    }
    void show() {
        System.out.println("k: " + k);
    }
}
class C extends B {
    C() {
        super(3, 2, 1);
        System.out.println("C is 1st.");
    }
    static void show(int a, int b, int c) {
        System.out.println("a = " + a);
        System.out.println("b = " + b);
        System.out.println("c = " + c);
    }
    void show(String msg) {
        System.out.println(msg + " i and j: " + i + " " + j);
    }
}
class MainClass {
    public static void main(String args[]) {
        C.show(1, 2, 3);
        C c = new C();
        c.show();
        c.show("ERROR!");
    }
}
```

Q3. Write output of the following java program.

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```
class Box {
    double width, height, depth;
    Box(double w, double h, double d) {
        width = w; height = h; depth = d;
    }
    double volume() { return width * height * depth; }
    public static void main(String args[]) {
        Box b1 = new Box(10, 10, 10);
        Box b2 = b1; Box b3 = new Box(20, 20, 20);
        System.out.println("B1=" + b1.volume() + ", B2=" + b2.volume() + ", B3=" + b3.volume());
        b1.width = b1.height = b1.depth = 30;
        b2.width = b2.height = b2.depth = 40;
        b3.width = b3.height = b3.depth = 50;
        System.out.println("B1=" + b1.volume() + ", B2=" + b2.volume() + ", B3=" + b3.volume());
    }
}
```

Term Test 2, CSE 233
Marks: 20, Time: 40min

Answer any two question including Q1

Q1. Define True/False. If false give the correct answer.

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- ✓ a) Inner class cannot be public.
- ✓ b) Since constructor does not have any return type, it cannot be overloaded.
- ✓ c) Protected members of a class cannot be accessed from non-subclass within same package.
- ✓ d) Abstract class cannot be inherited.
- ✓ e) All methods in an interface must be static.

Q2. Check the following java program carefully. Do you think there is any compilation error? If there is find the errors, correct it using line number and then write the output. For correcting you cannot insert new statement or delete a statement. If there is no error then write only output.

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```
01 abstract class A {
02     int i; System.out.println("Welcome 1"); }
03     A(int a) {
04         System.out.println("A is 3rd.");
05         i = a;
06     }
07     void show() { System.out.println("i : " + i); }
08     static { System.out.println("Static 2"); }
09 }
10 class B extends A {
11     int j; System.out.println("Welcome 2"); }
12     B(int a, int b) {
13         super(a);
14         System.out.println("B is 2nd.");
15         j = b; super.i = 20;
16     }
17     void show() { System.out.println("j : " + j); }
18     static { System.out.println("Static 3"); }
19 }
20 class C extends B {
21     C() {
22         super(2, 1);
23         System.out.println("C is 1st.");
24     }
25     static void show(int a, int b, int c) {
26         System.out.println("a = " + a);
27         System.out.println("b = " + b);
28         System.out.println("c = " + c);
29     }
30     void show(String msg) {
31         System.out.println(msg + " i and j: " + i + " " + super.j);
32     }
33     static { System.out.println("Static 1"); }
34 }
35 class MainClass {
36     public static void main(String args[]) {
37         C.show(1, 2, 3);
38         C c = new C();
39         c.show();
40         c.show("ERROR!");
41         A a = c;
42         a.show();
43     }
44 }
```

Handwritten notes on the right side of the code:

- a = 1
- b = 2
- c = 3
- Static 1
- Static 3

Term Test 3, CSE 233
Marks: 20, Time: 40min

Q1. Check the following java program carefully,

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```

01 class ExceptionExam {
02     public static void main(String args[]) {
03         System.out.println("Exam Started");
04         int a = args.length;
05         if (a == 0) a = 42;
06         System.out.println("a = " + a);
07         try {
08             if (a == 1) a = a / (a - a);
09             try {
10                 if (a == 2) { int c[] = {1}; c[42] = 99;
11                     System.out.println("c[42] = " + c[42]); }
12                 else if (a == 3)
13                     throw new NullPointerException("Null");
14                 else {
15                     System.out.println("Exam Ended Fast");
16                     return; }
17             } catch (ArrayIndexOutOfBoundsException e) {
18                 System.out.println("Divide by 0");
19             } finally { System.out.println("Un-reachable code"); }
20         } catch (ArithmeticException e) {
21             System.out.println("Array index out-of-bounds"); }
22         catch (Exception e) {
23             System.out.println("No Exception"); }
24         finally { System.out.println("Program Completed"); }
25         System.out.println("Exam Ended");
26     }
27 }

```

and write the output for the following commands:

- 1) java ExceptionExam
- 2) java ExceptionExam 0
- 3) java ExceptionExam 0 1
- 4) java ExceptionExam 0 1 2
- 5) java ExceptionExam 0 1 2 3

Q2. Check the following java program carefully. Do you think there is any compilation error? If there is the errors, correct it using line number and then write the output. For correcting you cannot insert new statement or delete a statement. If there is no error then write only output.

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```

01 public class ThreadExam implements Runnable {
02     public ThreadExam() {
03         Thread t = Thread.currentThread();
04         System.out.println("Current Thread: " + t);
05         t.setName("ExamThread"); t.setPriority(3);
06         System.out.println("Current Thread: " + t);
07         printer();
08     }
09     synchronized public void printer() {
10         Thread t = Thread.currentThread();
11         System.out.println("Printer: " + t);
12     }
13     public void run() {
14         Thread t = Thread.currentThread();
15         System.out.println("Current Thread: " + t);
16         try { Thread.sleep(1000); printer(); }
17         catch (InterruptedException e) { }
18         System.out.println("ThreadExam Exiting.");
19     }
20     public static void main (String args[]) {
21         Thread t = new Thread(new ThreadExam(), "MainThread");
22         System.out.println("Current Thread: " + t);
23         t.setPriority(7); t.start();
24         try { t.join(); }
25         catch (InterruptedException e) { }
26         System.out.println("ThreadMain Exiting.");
27     }
28 }

```

Shahjalal University of Science & Technology

Department of Computer Science and Engineering

2nd Year 2nd Semester Examination, 2016

Course: CSE 233, Course Title: Object Oriented Programming

Credit: 03 Full Marks: 40 Time: 40 minutes

Determine whether following the statements are true or false and explain your answer. (Any ten)

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- i. Java supports multiple inheritance.
- ii. A super class cannot have multiple sub class.
- iii. Signature and return type of a method must be same in case of method overloading.
- iv. In Java, main() function must be resided in a public class.
- v. Interpreter translates the entire program.
- vi. Java does not support implementation of multiple interface in a single class.
- vii. Wrapper classes allow primitive types to be accessed as objects. —
- viii. An abstract class can inherit an interface.
- ix. Interfaces may have instance variables.
- x. Finalize method is called when an object is out of scope.
- xi. "Person ob;" - here ob is assigned with an arbitrary pointer in memory. —
- xii. Super class variable cannot reference subclass object.

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Answer any five questions

- i. Why is Java called the "Platform Independent Programming Language"?
- ii. When can an object reference be cast to an interface reference?
- iii. What happens if a class partially implements an interface?
- iv. What is the purpose of declaring a class, variable or method as final?
- v. What happens when the constructor of a class is declared as "private"?
- vi. Differentiate between Call by Value and Call by Reference with appropriate example.
- vii. How does polymorphism enable you to program "in the general" rather than "in the specific"?

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Answer any four questions

- i. Can you override private or static method in Java?
- ii. Describe the circumstances in which an abstract method would be appropriate.
- iii. Differentiate among inner class, abstract class and interface.
- iv. What do you think the following fragment of code does?

```
MyClass c=new MyClass();
MyClass c2=c;
c=new MyClass();
```

- v. Where will be the most chance of the garbage collector being invoked?
- vi. What is output of the following code ?

```
1. class Garbage
2. {
3.     public static void main(String args[])
4.     {
5.         Garbage h = new Garbage();
6.         h.methodA();
7.     }
8.     Object methodA()
9.     {
10.        Object obj1 = new Object();
11.        Object [] obj2 = new Object[1];
12.        obj2[0] = obj1;
13.        obj1 = null;
14.        return obj2[0];
15.    }
16. }
```

```
class Parentclass {
    Parentclass(){
        System.out.println("Super Con");
    }
}
class Subclass extends Parentclass{
    Subclass(){
        System.out.println("Sub Con");
    }
    Subclass(int num){
        System.out.println("Con arg");
    }
    void display(){
        System.out.println("Hello");
    }
    public static void main(String args[]){
        Subclass obj= new Subclass();
        obj.display();
        Subclass obj2;
        obj2= new Subclass(10);
        obj2.display();
    }
}
```